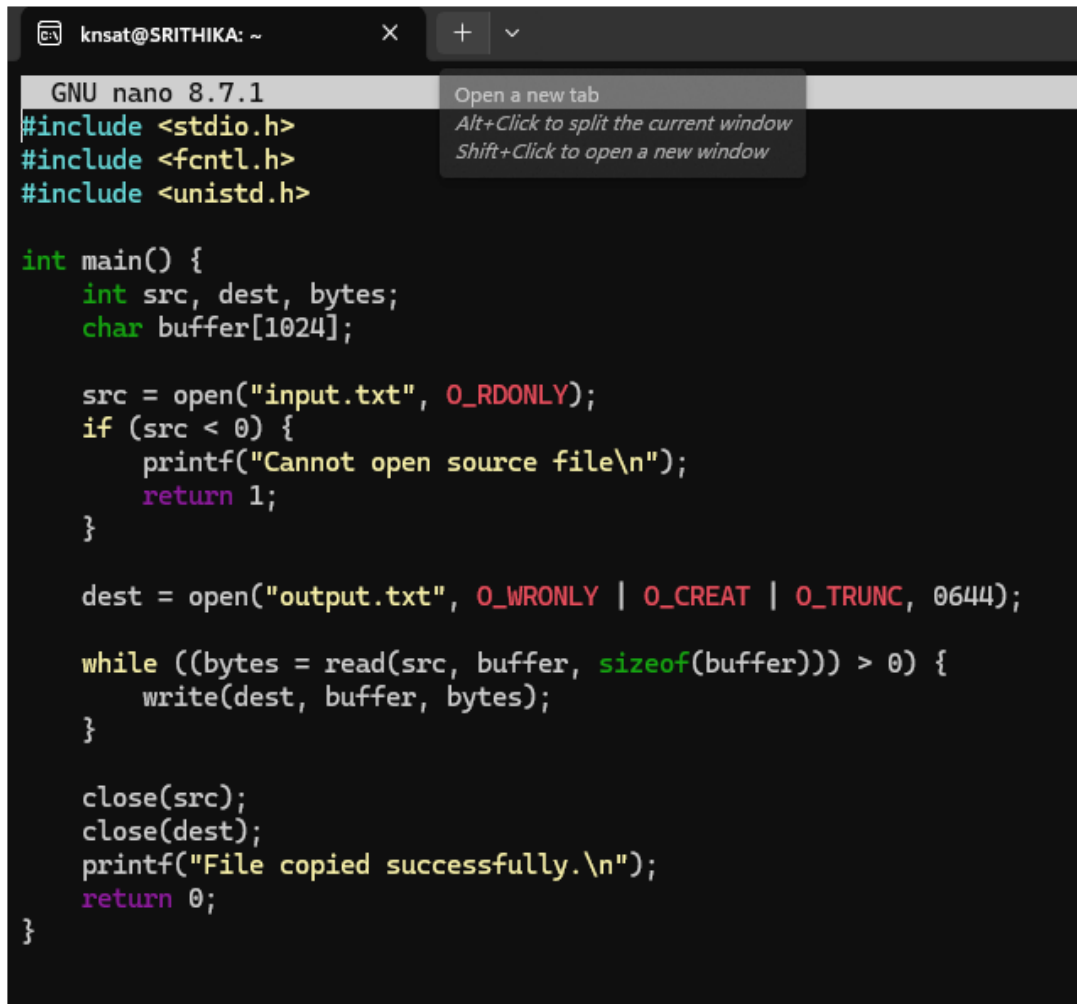


2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.



The screenshot shows a terminal window with the title `knsat@SRITHIKA: ~`. The terminal is running the GNU nano 8.7.1 text editor. The code being edited is a C program designed to copy the contents of `input.txt` to `output.txt`. The program includes `<stdio.h>`, `<fcntl.h>`, and `<unistd.h>`. It defines a `main()` function that declares `int` variables `src` and `dest`, and a `char` array `buffer` of size 1024. The program attempts to open `input.txt` in read-only mode (`O_RDONLY`). If this fails (`src < 0`), it prints an error message and returns 1. Otherwise, it opens `output.txt` in write-only mode with create and truncate flags (`O_WRONLY | O_CREAT | O_TRUNC`) and permissions of 0644. It then enters a `while` loop that reads data from `src` into `buffer` until `read` returns 0, and writes the data to `dest`. Finally, it closes both files, prints a success message, and returns 0. A context menu is visible over the top right of the editor, showing options: 'Open a new tab', 'Alt+Click to split the current window', and 'Shift+Click to open a new window'.

```
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int src, dest, bytes;
    char buffer[1024];

    src = open("input.txt", O_RDONLY);
    if (src < 0) {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

```
knsat@SRITHIKA: ~  
knsat@SRITHIKA:~$ echo "Hello, this is a test file for the OS lab assignment." > input.txt  
knsat@SRITHIKA:~$ nano copy.c  
knsat@SRITHIKA:~$ gcc copy.c -o copy  
knsat@SRITHIKA:~$ ./copy  
File copied successfully.  
knsat@SRITHIKA:~$ cat output.txt  
Hello, this is a test file for the OS lab assignment.  
knsat@SRITHIKA:~$ echo "This is sample text for strace." > sample.txt  
knsat@SRITHIKA:~$ strace cat sample.txt  
Command 'strace' not found, but can be installed with:  
sudo apt install strace  
knsat@SRITHIKA:~$ sudo apt install strace  
[sudo: authenticate] Password:  
Installing:  
strace  
  
Installing dependencies:  
libunwind8  
  
Summary:  
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 62  
Download size: 737 kB  
Space needed: 2660 kB / 1024 GB available  
  
Continue? [Y/n] Y  
Get:1 http://archive.ubuntu.com/ubuntu/resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]  
Get:2 http://archive.ubuntu.com/ubuntu/resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]  
Fetched 737 kB in 0s (1659 kB/s)  
Selecting previously unselected package libunwind8:amd64.  
(Reading database ... 40672 files and directories currently installed.)  
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...  
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...  
Selecting previously unselected package strace.  
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...  
Unpacking strace (6.19+ds-0ubuntu5) ...  
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...  
Setting up strace (6.19+ds-0ubuntu5) ...  
Processing triggers for man-db (2.13.1-1build1) ...
```